Name: _____

WORKSHEET

Sketching periodic functions

8	13	14	12	1	11	10	10	12	12	9	5	1	3	4	

8	14	7	8	6	1	14	12	13	2	11	10	

Jordan Williams

Rule

Given a periodic function $y = A \tan [B(x + C)] + D$

A is the **amplitude**

$\frac{2\pi}{B}$ is the **period**

C is the **phase shift**

D is the **vertical shift.**

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Questions

Given the periodic function $y = 3\sin\left[2\left(x - \frac{\pi}{2}\right)\right] + 1$:

1 State the amplitude.

2 State the period.

3 State the phase shift.

4 State the vertical shift.

5 Sketch this function.

Given the periodic function $y = 5\cos[4(x + \pi) - 2]$:

6 State the amplitude.

7 State the period.

8 State the phase shift.

9 State the vertical shift.

10 Sketch this function.

Sketch the following periodic functions.

11 $y = 3\tan[2(x + \pi)] - 1$

12 $y = 4\sin\left[3\left(x - \frac{\pi}{6}\right)\right] + 2$

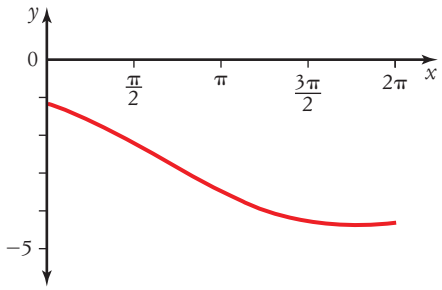
13 $y = 2\cos\left[\frac{1}{2}\left(x + \frac{\pi}{4}\right)\right] - 3$

14 $y = 5\tan\left[2\left(x - \frac{\pi}{2}\right)\right] + 1$

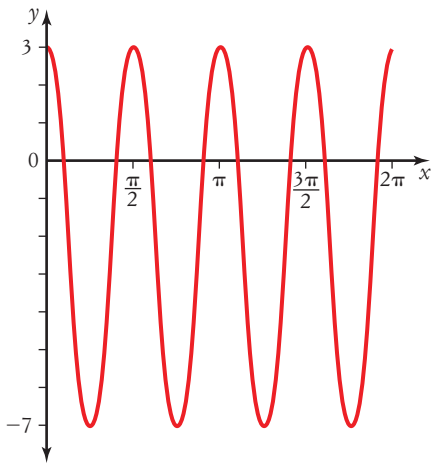
Solutions

A π to the left

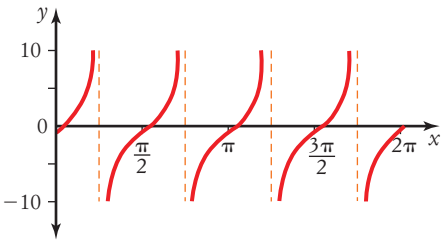
C



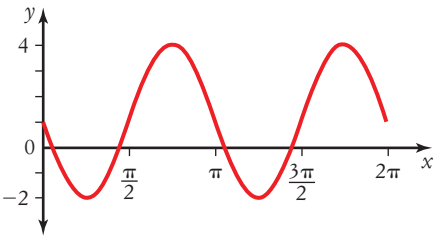
D



E

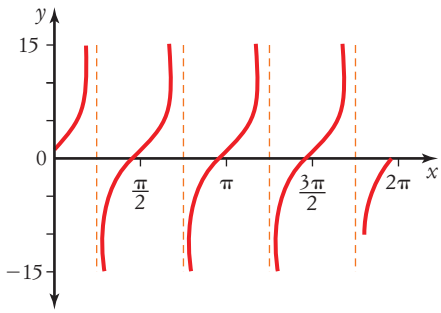


I



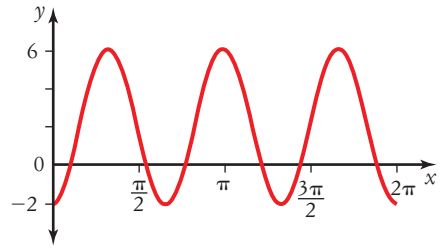
K π

L



N $\frac{\pi}{2}$ to the right

O



R 2 down

S 3

T 1 up

W $\frac{\pi}{2}$

Y 5

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